# RULE DETECTION

Figure Series: Movement, Rotation, Growth and Replacement

OAVS TGT | Reasoning (Non-Verbal) | Topic 13: fourth non-verbal topic

**Syllabus line (Reasoning, 30 Marks):** “The Non-Verbal Reasoning topics like Figure Formation, Dice, Triangle, **Rule Detection**, Images, Completion of Pattern, Mirror Images, Figure Matrix and Paper Folding, etc.”

**Built from:** OAVS TGT English 2019, English 2022, Mathematics 2024, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2). All 7 figure-series questions found in these papers are solved here with their original figures. Every rule was checked against every stage shown, and the practice figures were generated by program so each question has exactly one correct option.

## What is inside


| Ch. | Title | What you get |
|---|---|---|
| **1** | PYQ Analysis | Counts, rule types, priority, index of all 7 PYQs |
| **2** | Foundation | The element-tracking method and a catalogue of rules |
| **3** | Solved PYQs | Seven figure series, each with its rule and stage tables |
| **4** | Shortcuts, Traps and Scope | Speed methods and mistakes |
| **5** | Practice Set | 8 original figure series with a verified key |
| **6** | Quick Revision | One-page cheat sheet |

CHAPTER 1

# PYQ Analysis

Figure series: 7 questions in four of the five papers

| Paper | Q. numbers | Count | Rule type |
|---|---|---|---|
| **English 2019** | 36 | 1 | Parts of an outer and an inner polygon removed alternately |
| **English 2022** | none | 0 | — |
| **Mathematics 2024** | 75, 96 | 2 | Small symbol becomes big next stage (cycle of symbols); growing rows of cups |
| **Science 2024** | 79, 98 | 2 | Five symbols move round five positions |
| **Social Studies 2023** | 7, 26 | 2 | Arrow and flag rotate and swap rows; four symbols swap corners alternately |
| **Total** | | **7** | **Found in 4 of the 5 papers** |

| Rule type | PYQs | Priority |
|---|---|---|
| Symbols move between fixed positions (cycle or swaps) | 3 | ★★★ Must |
| Rotation of an element, with elements swapping places | 1 | ★★★ Must |
| Growth: one more line, more items each stage | 1 | ★★ High |
| Role swap: small and big symbols exchange, symbols cycle | 1 | ★★ High |
| Removal: parts erased stage by stage | 1 | ★★ High |
| *Expected, not yet asked:* shading changes, count progressions, odd figure out | 0 | ★ Expected |

**PYQ INSIGHT: Five of seven are ‘track each symbol’ questions.** Science 2024 Q79 and Q98 use a cycle through five positions (TL, TR, centre, BL, BR); SST 2023 Q26 swaps four corners. Write a table of positions for stages 1 and 2, find where each symbol moved, then apply the same moves to stage 4.

**PYQ INSIGHT: Two rules run at once in several PYQs** (SST 2023 Q7: the arrow turns anticlockwise while the flag turns clockwise, and they swap rows). Find the rule for each element separately.

## Index of the 7 PYQs

| Paper / Q | Rule in short | Ans | See |
|---|---|---|---|
| Eng 2019 Q36 | Outer then inner polygon loses a side, alternately | D | 3.1 |
| Maths 2024 Q75 | Small symbol at top-right becomes the big symbol at bottom-left next stage; new small symbol follows the cycle ○ P △ ∞ □ | A | 3.2 |
| Maths 2024 Q96 | Stage n has n rows with n, n−1, …, 1 cups, left-aligned | C | 3.3 |
| Science 2024 Q79 | Five symbols move one step round: TL←TR←BR←BL←C←TL | 1 | 3.4 |
| Science 2024 Q98 | Five symbols move: TL←TR←C←BR←BL←TL | 1 | 3.5 |
| SST 2023 Q7 | Arrow turns anticlockwise, flag clockwise; they swap rows | 4 | 3.6 |
| SST 2023 Q26 | Alternately: turn 180° (diagonal swap), then swap left and right | 2 | 3.7 |

CHAPTER 2

# Foundation

How to find the rule

## 2.1 Method: track each element

| Step | Do this |
|---|---|
| **1** | List the **elements** in the figure (symbols, lines, shaded parts) and the **features** that can change: position, direction, size, shading, number. |
| **2** | Compare stage 1 with stage 2 element by element. Write what happened to each (moved, turned, grew, vanished, replaced). |
| **3** | Check stage 2 with stage 3 to confirm. If the change differs, the rule alternates or grows. |
| **4** | Apply the confirmed rule to stage 4 to get stage 5. Check every element of the answer. |
| **5** | Eliminate options one feature at a time: position first, direction second, shading last. |

## 2.2 Catalogue of rules

| Rule | What it looks like | PYQ |
|---|---|---|
| **Cyclic move among positions** | Every symbol shifts to the next position in a fixed loop (five positions or four corners) | Science Q79, Q98 |
| **Swap** | Two or four symbols exchange places (left-right, up-down, diagonal) | SST Q26 |
| **Rotation** | An element turns by a fixed angle each stage (90° or 45°, clockwise or anticlockwise) | SST Q7 |
| **Alternating rule** | Rule 1 on odd steps, rule 2 on even steps | SST Q26, Eng 2019 Q36 |
| **Growth** | One more line, one more part, one more row each stage | Maths Q96 |
| **Removal** | A side or part is erased each stage | Eng 2019 Q36 |
| **Replacement / role swap** | A symbol is replaced by the next in a cycle; small becomes big | Maths Q75 |
| **Shading** | A shaded part moves or grows (expected) | — |

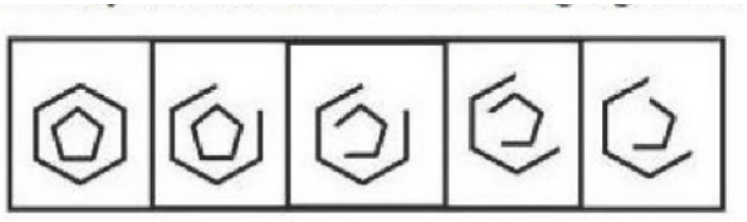
**SHORTCUT: Position coding:** name the five positions TL, TR, C, BL, BR (centre). Write each stage as a row of five symbols in that order. The move shows up as a pattern in the table.

CHAPTER 3

# Solved PYQs

Seven figure series with their rules

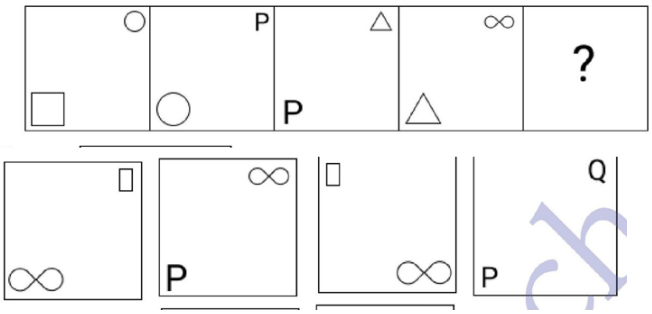
## 3.1 English 2019 Q36: sides removed alternately



**1.** Each stage shows an outer polygon (a hexagon) with an inner pentagon.
**2.** Stage 1 to 2: the outer hexagon loses a side; the pentagon is intact. Stage 2 to 3: the pentagon loses a side. Stage 3 to 4: the outer polygon loses another. Stage 4 to 5: the inner polygon loses another.
**3.** So the number of sides of the outer and the inner polygon **reduce in alternate stages** (counts read from the figure: outer 6, 5, 5, 4, 4; inner 5, 5, 4, 4, 3).
**Answer: D: number of sides of an outer and inner polygon reducing in each alternate stage**

| Stage | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| **Outer sides** | 6 | 5 | 5 | 4 | 4 |
| **Inner sides** | 5 | 5 | 4 | 4 | 3 |

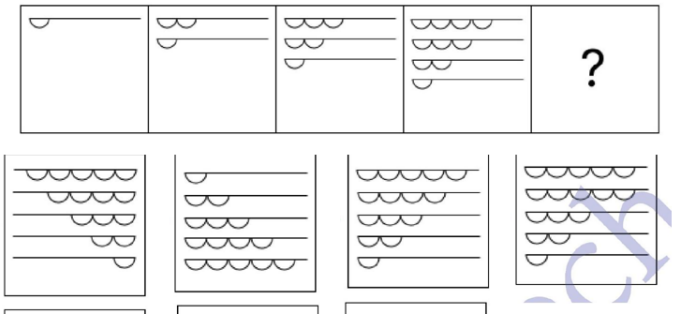
## 3.2 Maths 2024 Q75: small symbol becomes the big one




**1.** Each box has a **small symbol at top-right** and a **big symbol at bottom-left**.
**2.** The small symbol of one stage becomes the big symbol of the next: ○ (stage 1 small) is big in stage 2; P is big in stage 3; △ is big in stage 4. So in stage 5 the big symbol is the stage 4 small symbol, ∞.
**3.** The small symbols follow a cycle: ○, P, △, ∞ and then the first big symbol, □ (the symbols are □ ○ P △ ∞ repeating). So the new small symbol is □.
**Answer: A (small □ at top-right, big ∞ at bottom-left)**

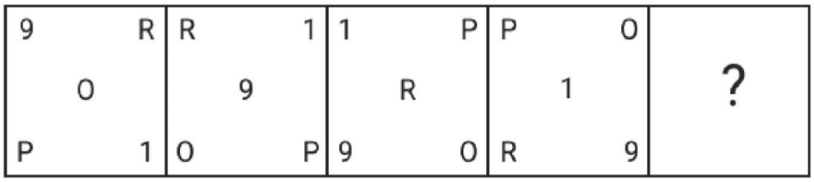
| Stage | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| **Small (top-right)** | ○ | P | △ | ∞ | □ |
| **Big (bottom-left)** | □ | ○ | P | △ | ∞ |

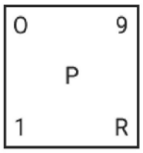
## 3.3 Maths 2024 Q96: growing rows of cups




**1.** Stage 1: 1 line with 1 cup. Stage 2: 2 lines, with 2 cups on the first and 1 on the second. Stage 3: 3, 2, 1. Stage 4: 4, 3, 2, 1.
**2.** Each stage adds a new line at the bottom with 1 cup, and adds one cup to every earlier line. The cups hang from the **left** end of each line.
**3.** Stage 5 therefore has 5 lines with 5, 4, 3, 2, 1 cups, left-aligned. Option A has the staircase on the right and option B has the counts in the reverse order, so only the left-aligned 5, 4, 3, 2, 1 figure fits.
**Answer: C**

## 3.4 Science 2024 Q79: five symbols, five positions

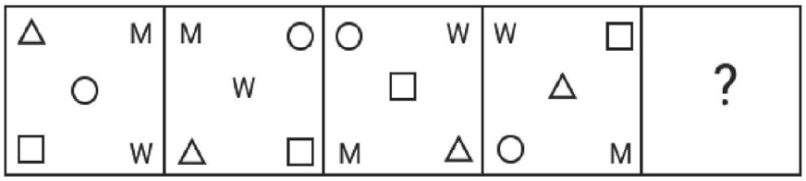


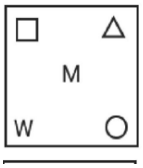




**1.** Positions in each box: TL, TR, C (centre), BL, BR. Stage 1: 9, R, O, P, 1. Stage 2: R, 1, 9, O, P.
**2.** Compare stage 1 and 2: R moved from TR to TL; 1 from BR to TR; P from BL to BR; O from C to BL; 9 from TL to C. So **new TL = old TR, new TR = old BR, new BR = old BL, new BL = old C, new C = old TL**.
**3.** This rule also turns stage 2 into stage 3 (1, P, R, 9, O) and stage 3 into stage 4 (P, O, 1, R, 9). Apply it to stage 4: TL = O, TR = 9, BR = R, BL = 1, C = P.
**Answer: Option 1 (O 9 / P / 1 R)**

| Stage | TL | TR | C | BL | BR |
|---|---|---|---|---|---|
| **1** | 9 | R | O | P | 1 |
| **2** | R | 1 | 9 | O | P |
| **3** | 1 | P | R | 9 | O |
| **4** | P | O | 1 | R | 9 |
| **5 (answer)** | O | 9 | P | 1 | R |

## 3.5 Science 2024 Q98: five symbols, a different loop

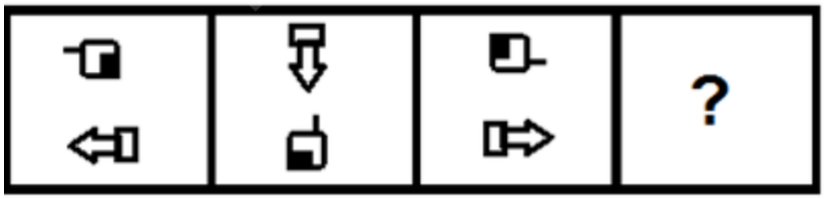



**1.** Stage 1: △ (TL), M (TR), ○ (C), □ (BL), W (BR). Stage 2: M, ○, W, △, □.
**2.** Moves: M from TR to TL; ○ from C to TR; W from BR to C; □ from BL to BR; △ from TL to BL. So **new TL = old TR, new TR = old C, new C = old BR, new BR = old BL, new BL = old TL**.
**3.** Check on stages 2 to 3 and 3 to 4 (it holds). Stage 4: W, □, △, ○, M gives stage 5: TL = □, TR = △, C = M, BR = ○, BL = W.
**Answer: Option 1 (□ △ / M / W ○)**

| Stage | TL | TR | C | BL | BR |
|---|---|---|---|---|---|
| **1** | △ | M | ○ | □ | W |
| **2** | M | ○ | W | △ | □ |
| **3** | ○ | W | □ | M | △ |
| **4** | W | □ | △ | ○ | M |
| **5 (answer)** | □ | △ | M | W | ○ |

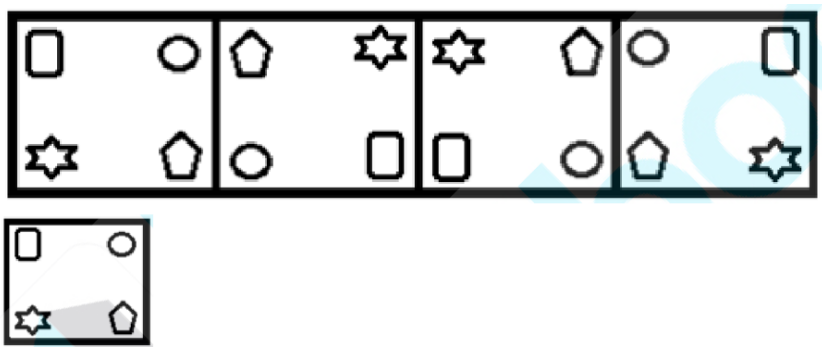
## 3.6 SST 2023 Q7: turning arrow and flag, swapping rows






**1.** Each box holds an arrow and a flag-like symbol (a box with a stem and one shaded quarter). They **swap rows** each stage: arrow bottom, top, bottom, so the arrow is on top in stage 4.
**2.** **Arrow:** left, down, right: it turns 90° anticlockwise each stage, so stage 4 points up.
**3.** **Flag:** stem left (shaded bottom-right), stem up (shaded bottom-left), stem right (shaded top-left): it turns 90° clockwise each stage, so stage 4 has the stem down and the shading at top-right, and it sits in the bottom row.
**4.** Option 4 matches: arrow up on top; flag with stem down and top-right shading at the bottom. Option 2 has the rows not swapped; option 3 has the flag unrotated (stem up).
**Answer: Option 4**

## 3.7 SST 2023 Q26: four symbols, alternating swaps



**1.** Symbols: R = rounded rectangle, C = circle, S = star, P = pentagon. Stage 1: R (TL), C (TR), S (BL), P (BR). Stage 2: P, S, C, R: every symbol went to the opposite corner (a turn of 180°, or a diagonal swap).
**2.** Stage 2 to 3: P and S swap in the top row, C and R swap in the bottom row: **left and right swap**. Stage 3 to 4: diagonal swap again. So the rule alternates: diagonal swap, left-right swap, diagonal swap, ...
**3.** Stage 4 to 5 is a left-right swap: stage 4 (C, R, P, S) becomes R, C, S, P, the same as stage 1.
**Answer: Option 2**

| Stage | TL | TR | BL | BR |
|---|---|---|---|---|
| **1** | R | C | S | P |
| **2** | P | S | C | R |
| **3** | S | P | R | C |
| **4** | C | R | P | S |
| **5 (answer)** | R | C | S | P |

CHAPTER 4

# Shortcuts, Traps and Scope

Speed methods and mistakes

- **Write a position table** (TL, TR, C, BL, BR) for each stage; the loop is obvious in 20 seconds.
- **Test the rule on two transitions** (1→2 and 2→3) before applying it to 4→5.
- **Alternating rules:** if 1→2 and 2→3 differ but 1→2 and 3→4 agree, the rule alternates.
- **Separate elements:** an arrow and a flag can follow different rules (SST 2023 Q7).
- **Eliminate by position first,** then by direction or shading.

| Trap | How to avoid it |
|---|---|
| **Applying the stage 1→2 rule only** | Check it on stage 2→3; the rule may alternate. |
| **Mixing up clockwise and anticlockwise** | Mark the direction on the first two stages and keep it. |
| **Forgetting that the second element also moves** | Track every element, not just the obvious one. |
| **Choosing an option with the right symbols in the wrong places** | Compare each position separately. |
| **Ignoring shading** | In SST 2023 Q7 the shaded quarter turns with the flag. |

## Scope

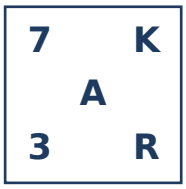
| Related PYQs | Where |
|---|---|
| Mirror images (Eng 2019 Q34, Q46; Eng 2022 Q35; Maths Q77, Q79, Q93; Science Q97; SST Q4) | Mirror Images (coming) |
| Completion of Pattern (Eng 2019 Q56); Figure Matrix (Eng 2019 Q53; Eng 2022 Q47) | Their own PDFs (coming) |

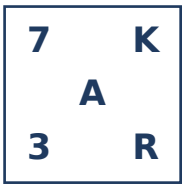
CHAPTER 5

# Practice Set

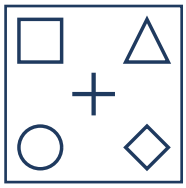
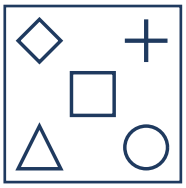
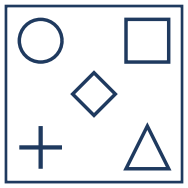
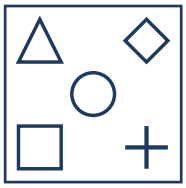
8 original figure series; answers on the next page (generated and verified by program)

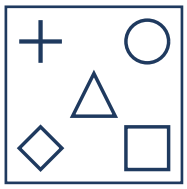
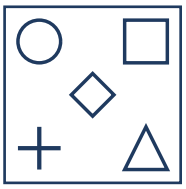
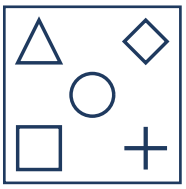
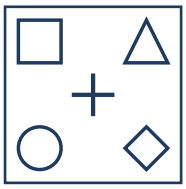
**Q1.** Find the figure that continues the series (five positions).

| 1 | 2 | 3 | 4 | ? |
|---|---|---|---|---|
|  |  |  |  |  |

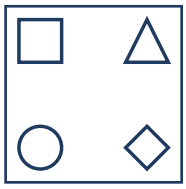
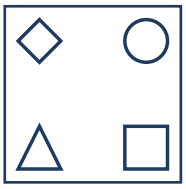
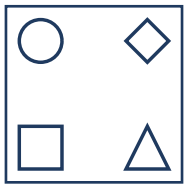
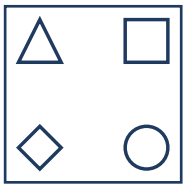
| (A) | (B) | (C) | (D) |
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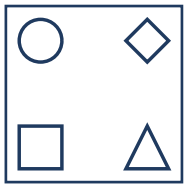
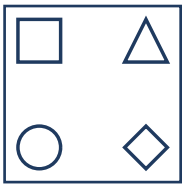
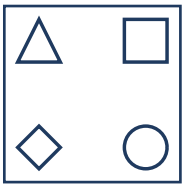
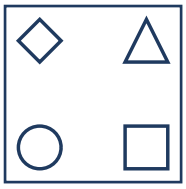
**Q2.** Find the figure that continues the series (five positions).

| 1 | 2 | 3 | 4 | ? |
|---|---|---|---|---|
|  |  |  |  |  |

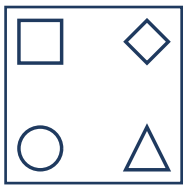
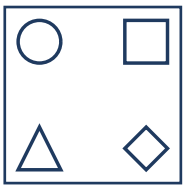
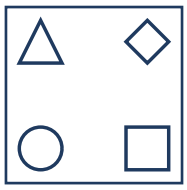
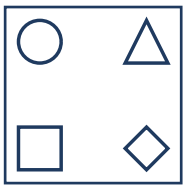
| (A) | (B) | (C) | (D) |
|---|---|---|---|
|  |  |  |  |

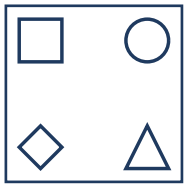
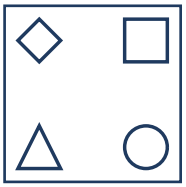
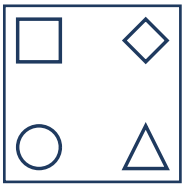
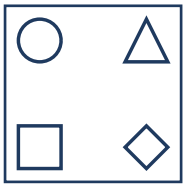
**Q3.** Find the figure that continues the series (four corners; the rule alternates).

| 1 | 2 | 3 | 4 | ? |
|---|---|---|---|---|
|  |  |  |  |  |

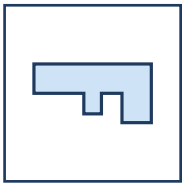
| (A) | (B) | (C) | (D) |
|---|---|---|---|
|  |  |  |  |

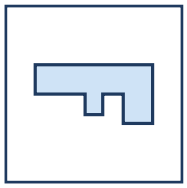
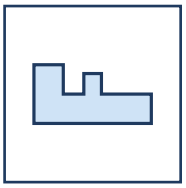
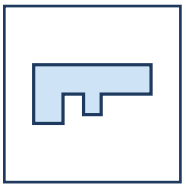
**Q4.** Find the figure that continues the series (four corners; the rule alternates).

| 1 | 2 | 3 | 4 | ? |
|---|---|---|---|---|
|  |  |  |  |  |

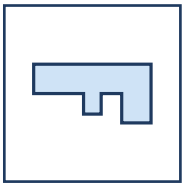
| (A) | (B) | (C) | (D) |
|---|---|---|---|
|  |  |  |  |

**Q5.** Find the figure that continues the series.

| 1 | 2 | 3 | ? |
|---|---|---|---|
|  |  |  |  |

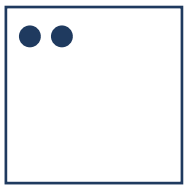
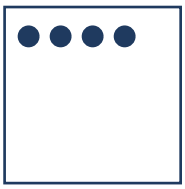
| (A) | (B) | (C) | (D) |
|---|---|---|---|
|  |  |  |  |

**Q6.** Find the figure that continues the series.

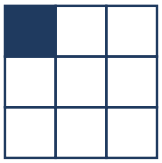
| 1 | 2 | 3 | ? |
|---|---|---|---|
|  |  |  |  |

| (A) | (B) | (C) | (D) |
|---|---|---|---|
|  |  |  |  |

**Q7.** The figures show 2, 4, 7 and 11 dots. How many dots will the next figure have?

|  |  |  |  |
|---|---|---|---|
| **1** | **2** | **3** | **4** |

**Q8.** Find the figure that continues the series (shading grows).

|  |  |  |  |  |
|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **?** |

|  |  |  |  |
|---|---|---|---|
| **(A)** | **(B)** | **(C)** | **(D)** |

## Answer key

| Q | Answer | Rule |
|---|---|---|
| **1** | (D) | Each symbol moves one step round a five-position loop: BL ← C, BR ← BL, TL ← BR, TR ← TL, C ← TR (new position ← old position). |
| **2** | (A) | Each symbol moves one step round a five-position loop: TR ← C, BL ← TR, BR ← BL, TL ← BR, C ← TL (new position ← old position). |
| **3** | (B) | Two operations alternate on the four corners (stage 1→2, 3→4 use one; 2→3, 4→5 use the other). |
| **4** | (C) | Two operations alternate on the four corners (stage 1→2, 3→4 use one; 2→3, 4→5 use the other). |
| **5** | (B) | The shape turns 90° clockwise each stage; the answer is turned 270° and not flipped. |
| **6** | (A) | The shape turns 45° clockwise each stage; the answer is turned 135° and not flipped. |
| **7** | 16 | Differences +2, +3, +4, then +5: 11 + 5 = 16. |
| **8** | (A) | One more border square is shaded each stage, clockwise from the top-left: stage 5 has 5 squares shaded. |

CHAPTER 6

# Quick Revision

One page for the last day

| Rule | How to spot | PYQ |
|---|---|---|
| **Cycle among positions** | Same symbols in every stage, each in a new place | Science Q79, Q98 |
| **Alternating swaps** | Stage 1→2 differs from 2→3, but 1→2 matches 3→4 | SST Q26 |
| **Rotation + swap** | Arrow and flag turn in opposite directions and trade rows | SST Q7 |
| **Role swap** | Small becomes big next stage; symbols cycle | Maths Q75 |
| **Growth** | One more line and one more cup each stage | Maths Q96 |
| **Removal** | A side is erased alternately outer and inner | Eng 2019 Q36 |

**ROUTINE:** **Exam-day routine:** table the elements; rule from 1→2; confirm on 2→3; apply to 4→5; eliminate options by position, direction, shading. 40 seconds per question. Next in the syllabus order: Images, Completion of Pattern, Mirror Images, Figure Matrix, Paper Folding.